

EDUGENIE-AI

Performance Testing

Project Name: EduGenie-AI

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Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Apache JMeter / Postman
Type of Testing	Load Testing, Stress Testing, Spike Testing
Target Module / API	EduGenie-AI web application and API endpoints
Test Environment	Local development server
Test Date	Not included

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Normal student requests	10	60	Successful AI response
2	Concurrent student requests	25	60	Stable response without critical errors
3	High-load requests	50	60	Application remains available
4	Short spike of requests	75	30	Requests are handled without application failure

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Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 5 seconds	To be recorded	Pass / Fail	Record after test execution
2	Response Time (Max)	< 5 seconds	To be recorded	Pass / Fail	Record after test execution
3	Throughput (Requests/sec)	> 10	To be recorded	Pass / Fail	Record after test execution
4	Error Rate	< 1%	To be recorded	Pass / Fail	Record after test execution
5	CPU Utilization	< 80%	To be recorded	Pass / Fail	Record after test execution
6	Memory Utilization	< 80%	To be recorded	Pass / Fail	Record after test execution

Step 4: Observations & Analysis

Area	Observation
Key Findings	The test results should show how the EduGenie-AI application performs when multiple student requests are processed.
Bottlenecks Identified	Potential bottlenecks include AI API response time, request processing time, and high concurrent load.
Optimization Steps Taken	Optimize request handling, reduce unnecessary processing, improve error handling, and monitor API response time.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Postman results, response graphs).

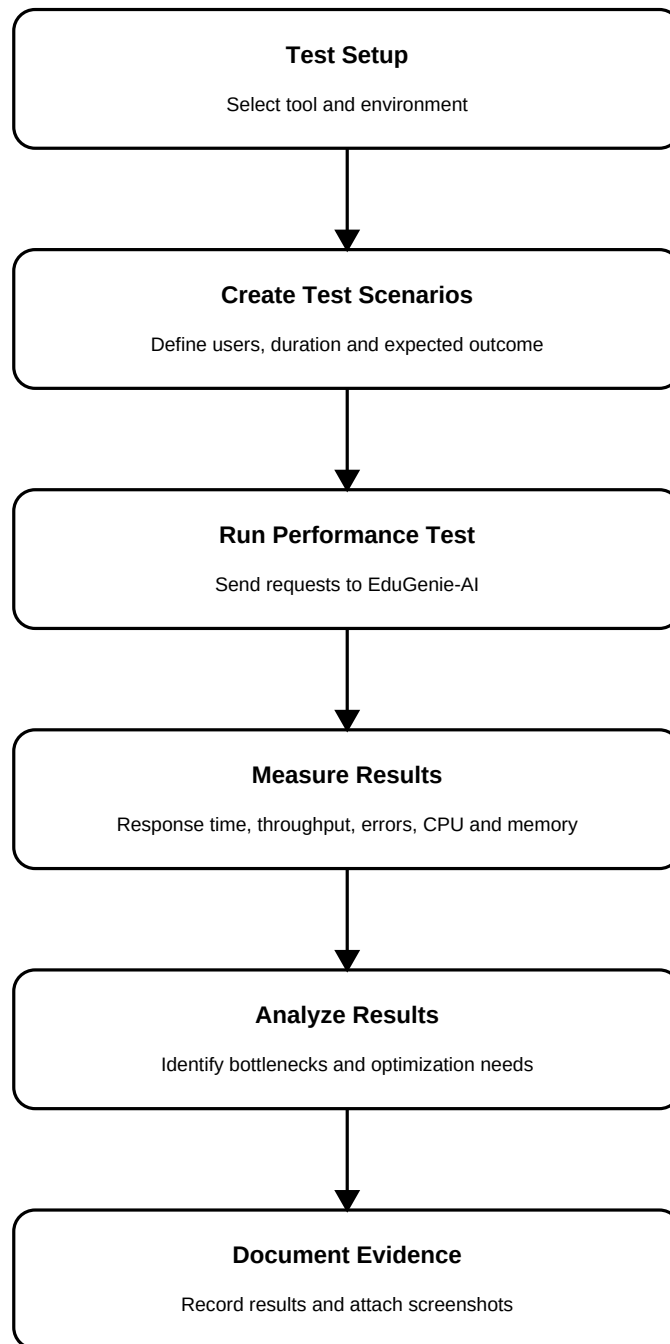
[Insert Performance Test Screenshot 1 here]

[Insert Performance Test Screenshot 2 here]

EduGenie-AI Performance Testing Flow

Flow Diagram

The flow below presents the performance-testing process for the EduGenie-AI application, from test setup through execution, measurement, analysis, and evidence.



Flow: Test Setup → Create Test Scenarios → Run Performance Test → Measure Results → Analyze Results → Document Evidence